Connor L. Brown, PhD

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Research Appointments

Post-Doctoral Researcher *Jan 2025 – present*
Dept. of Chemical and Biosciences, Aalborg University Adviser: Mads Albertsen

Post-Doctoral Researcher *Jan 2024 – Jan 2025*
Dept. of Civil and Environmental Engineering, Virginia Tech Adviser: Amy Pruden

Graduate Research Assistant *2018 – Dec 2023*
Dept. of Civil and Environmental Engineering, Virginia Tech Adviser: Amy Pruden

Undergraduate Research Assistant *2016 – 2018*
Dept. of Civil and Environmental Engineering, Virginia Tech

Education

Ph.D., Genetics, Bioinformatics, and Computational Biology (GBCB) *Aug 2019 – Dec 2023*
Virginia Tech Advisor: Amy Pruden
Dissertation: Bioinformatic Analysis of Wastewater Metagenomes Reveals Microbial Ecological and Evolutionary Phenomena Underlying Associations of Antibiotic Resistance with Antibiotic Use

M.S., Biochemistry *May 2019*
Virginia Tech Advisor: Biswarup Mukhopadhyay
Thesis: Secretion and environmental biochemistry of *Legionella pneumophila* in corrosive water

B.S., Biochemistry *May 2018*
Virginia Tech

Selected Honors and Awards

2024 Outstanding Dissertation Award, GBCB
2022 Outstanding Doctoral Student in Interdisciplinary Programs, Virginia Tech
2021 Outstanding Interdisciplinary Doctoral Student, GBCB
2018 Commendation for Excellence in Undergraduate Research
2017 ATC Creativity Student Grant
2016 Inaugural Fralin Undergraduate Research Fellowship

Professional Service

2021–2023 GBCB Senator to the Graduate and Professional Student Senate
2022 Co-chair, Virginia Tech 38th Annual Graduate and Professional Student Senate Research Symposium
2019–2020* Ximbio, Campus Representative for Life Science Reagent Non-Profit
2018–present American Society for Biochemistry, Student Member
2016–present American Society for Microbiology, Student Member
*Position ended due to COVID-19

Teaching Service

Graduate Teaching Assistant *Aug–Dec 2019*
Virginia Tech BCHM4115 (First of two for-majors Biochemistry courses)

Teaching Assistant *May 2022*
Student-led metagenomics workshop

Guest Lecturer *Jan 2024*

Virginia Tech Transdisciplinary Research Course

Peer Review Service

Frequent Peer Reviewer for *Water Research*, *Nature Communications*, *Microbiome*/*Environmental Microbiome*, *PNAS*, *Environmental Science: Nano*. Reviewing Editor for *Microbiome*.

Contributions to Externally Funded Research

Community Prevalence and Transmission Pathways of Multidrug Resistant Carbapenemase-producing Enterobacteriaceae *2024*
National Institute of Health R01 (2024–2029 est.) P.I.: Latania Logan (Emory University); co-PI: Amy Pruden
Personal contribution: Contributed bioinformatics expertise towards the development of source tracking methodology, selection of culture methods, and multiple rounds of editing.

Quantifying Wastewater Sources of Antibiotic Resistance to Aquatic and Soil Environments and Associated Human Health Risks ($2,374,575) *2024*
Environmental Protection Agency (2024–2027 est.) PI: Lola Olabode (Water Research Foundation)
Personal contribution: Contributed bioinformatics expertise and proposal of novel computational methods to address multivariate antibiotic susceptibility profiles of pathogens predicted solely from metagenomic data.

Advancing Comprehensive Wastewater Surveillance of Endemic and Emerging Pathogens Across Multiple Microbial Domains through Strain-resolved Metagenomics ($584,968) *2022*
Centers for Disease Control (2022–2024) PI: Amy Pruden (VT); Co-PIs: Liqing Zhang (VT), Peter Vikesland (VT)
Personal contribution: Lead author for the Pruden lab. Contributed to proposal development, project management, and intellectual development of lab methods and software to recommend a protocol to the CDC National Wastewater Surveillance System (NWSS) capturing multiple pathogens and ARGs.

WRF 4961: The Use of NGS Technologies and Metagenomics Approaches to Evaluate Water and Wastewater Quality Monitoring and Treatment Technologies ($490,015) *2020*
Water Research Foundation (2020–2023) PI: Amy Pruden (VT); Co-PI: Emily Garner (WVU)
Personal contribution: Contributed to an extensive guidance document for US water utilities on NGS for water and wastewater quality assessment. Primary author of the chapter on metagenomic assembly algorithms.

Frameworks: Developing Cyberinfrastructure for Waterborne Antibiotic Resistance Risk Surveillance (CI4-WARS) ($1,299,036) *2020*
National Science Foundation (2020–2024) PI: Liqing Zhang (VT); Co-PIs: Amy Pruden (VT), Peter Vikesland (VT), Ali Butt (VT)
Personal contribution: Led major sampling efforts for a prototype metagenomics-based environmental monitoring program at three full-scale wastewater treatment plants. Oversaw collection of >1,000 wastewater samples and generation of 11×10^{12} basepairs of Illumina and nanopore metagenomic sequences.

Pending or Unfunded Research Proposals

BIOSmartSewers: harmonizing next generation public health monitoring and wastewater treatment via biofilm-based biological sensors (12,487,026 kr / $1,354,973) *2026*
Novo Nordisk Foundation Emerging Investigator in Biotechnology. P.I.: Connor L. Brown (Aalborg University). Not funded (2026–2029).

Mapping the spatial distribution of antimicrobial resistance genes in hospital- and community-associated pathogens at sewershed-scale ($3,896,733) *2025*
National Institutes of Health. P.I.: Amy Pruden (Virginia Tech). Not funded (2026–2029).
Personal contribution: DNA sequencing and bioinformatics consultant. Developed experimental design, contributed preliminary data, proposed bioinformatic methods, figures, and writing.

ARsourcetracker: metagenomics based source tracking of environmental antibiotic resistance ($1.5M) *2025*
National Science Foundation. P.I.: Liqing Zhang (Virginia Tech). Not funded (2026–2029).
Personal contribution: DNA sequencing and bioinformatics consultant. Developed experimental design, contributed preliminary data, proposed bioinformatic methods, figures, and writing.

Charting pathways and barriers to novel antibiotic resistance gene emergence and global dissemination ($270,000) *2024*
National Science Foundation Post-Doctoral Fellowship. Not funded (2025–2028).
Sponsoring scientists: Simon Roux (Joint Genome Institute); Eduardo Rocha (Pasteur Institute); Mads Albertsen (Aalborg University).

ARtracker: A foundational cyberinfrastructure for One Health source tracking of antibiotic resistance using metagenomic and whole genome sequencing data ($1,300,000) *2023*
PI: Liqing Zhang. Not funded (2024–2027).
Personal contribution: Lead author for the Pruden lab. Contributed bioinformatics expertise towards source tracking methodology, culture methods selection, and multiple rounds of editing.

IntBio: A Predictive, Biologically Aware Framework for Horizontal Gene Transfer Dynamics in Wastewater ($1,742,708) *2023*
National Science Foundation. PIs: Allison Lopatkin (U. Rochester), Amy Pruden (VT), Lenwood Heath (VT), Peter Vikesland (VT), Liqing Zhang (VT), Jillian Pope (FAMU). Not funded (2024–2027).
Personal contribution: Lead author for the Pruden lab. Conducted experiments, contributed preliminary results, developed research objectives, writing, literature review, and laboratory experiment design.

NLM: Developing a comprehensive data and tool resource for identification of novel antibiotic resistance genes ($1,494,786) *2021*
National Institutes of Health. PIs: Liqing Zhang (VT), Lifu Huang (VT), Amy Pruden (VT), Peter Vikesland (VT). Not funded (2022–2025).

Edge: ARGnet: Developing a comprehensive data and tool resource for identification of novel antibiotic resistance genes ($1,300,000) *2021, 2023*
National Science Foundation. PIs: Liqing Zhang (VT), Anne Brown (VT), Lenwood Heath (VT), Amy Pruden (VT), Peter Vikesland (VT). Not funded (2022–2026).
Personal contribution: Primary author of the proposal. Led development of a prototype online resource for identification of putative antibiotic resistance genes, including protein structural modeling, large-scale phylogenomics, and integration with deep learning models.

Publications

Peer-Reviewed Journal Articles in Print (26)

1. **Brown, C.L.**, Rumi, M.A., McDaniel, L., Maile-Moskowitz, A., Sein, J., Choi, M., Hindi, F., Mullet, J., Emon, M., Moumi, N.A., Blair, M.F., Rao, J., Baffoe-Bonnie, A., Vikesland, P., Pruden, A., Zhang, L. Metagenomics disentangles ecological and epidemiological associations between community antibiotic use and sewage antibiotic resistance indicators. *Nat Commun* https://doi.org/10.1038/s41467-026-78142-3

2. **Brown, C.L.**, Maile-Moskowitz, A., Lopatkin, A.J., Xia, K., Zhang, L., Davis, B., Vikesland, P., Pruden, A. Selection and horizontal gene transfer underlie microdiversity-level heterogeneity in resistance gene fate during wastewater treatment. *Nat Commun* **15**, 5412 (2024). https://doi.org/10.1038/s41467-024-49742-8
3. **Brown, C.L.**, Mullet, J., Hindi, F., Stoll, J., Gupta, S., Choi, M., Vikesland, P., Keenum, I., Pruden, A., Zhang, L. mobileOG-db: a manually curated reference database of proteins mediating the life cycle of bacterial mobile genetic elements. *Appl. Env. Micro.* https://doi.org/10.1128/aem.00991-22
4. **Brown, C.L.**, Keenum, I., Dai, D., Vikesland, P., Pruden, A., Zhang, L. (2021). Critical evaluation of short, long, and hybrid assembly for contextual analysis of antibiotic resistance genes in complex environmental metagenomes. *Scientific Reports* **11**(1): 3753. https://doi.org/10.1038/s41598-021-83081-8
5. **Brown, C.L.**, Garner, E., Jospin, G., Coil, D.A., Schwake, D.O., Eisen, J.A., Mukhopadhyay, B., & Pruden, A.J. (2020). Whole genome sequence analysis reveals the broad distribution of the RtxA type 1 secretion system and four novel putative type 1 secretion systems throughout the *Legionella* genus. *PLOS ONE*, **15**(1), e0223033. https://doi.org/10.1371/journal.pone.0223033
6. **Brown, C.L.***, Huang, E., Garcia, J., Brown, A. National correspondence test reveals distinct patterns of transfer student perceptions by researchers of differing seniority at R1 universities. *Accepted, Discover Education.*
7. Mullet, J., Zhang, L., Pruden, A., **Brown, C.L.*** Phage-plasmid hybrids are found across diverse ecosystems and encode niche-specific functional traits. *PLoS ONE.* https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0350027
8. Choi, J.M.; **Brown, C.L.**; Pruden, A.; Zhang, L. VRGAN: A Deep Generative Framework for Unsupervised Anomaly Detection in Multivariate Time-Series Data. *Appl. Sci.* 2026, 16, 9556. https://doi.org/10.3390/app16199556
9. Maile-Moskowitz, A., **Brown, C.L.**, Logan, L., Xia, K., Pruden, A., Vikesland, P. Impact of hospital wastewater on simulated activated sludge treatment and the antibiotic resistome. *Environ. Sci.: Water Res. Technol.* 2026; 12 (9): 2975–2990. https://doi.org/10.1039/d5ew01122a
10. Chong, E.C., Pham, T., Carey, M., Wee, B., Taouk, M., Fernandez-Favieres, F., Moore, E., Lim, C., **Brown, C.L.**, Williamson, D., Opatowski, L., Outterson, K., Mukiri, K., Essack, S.Y., Brisse, S., Grad, Y., Baker, K. The 2024 Antimicrobial Resistance—Genomes, Big Data and Emerging Technologies Conference: Summary of proceedings. *NPJ Antimicrobials and Resistance.* https://www.nature.com/articles/s44259-024-00058-z
11. Maile-Moskowitz, A., **Brown, C.L.**, Rumi, M.A., Sein, J., Choi, M., Hindi, F., Mullet, J., Emon, M., Moumi, N.A., Blair, M.F., Rao, J., Baffoe-Bonnie, A., Zhang, L., Pruden, A., Vikesland, P. Can wastewater-based surveillance relate antimicrobial use to resistance patterns? Metagenomic analysis of two neighboring treatment plants. *In press, Nature Antimicrobials and Resistance.* https://www.researchsquare.com/article/rs-4420801/v1
12. Moumi, N.A., Ahmed, S., **Brown, C.L.**, Pruden, A., Zhang, L. ARGContextProfiler: extracting and scoring the genomic contexts of antibiotic resistance genes using assembly graphs. *Frontiers in Microbiology.* https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2025.1604461/full
13. Muhit, E., Fei Cheung, Y., Stoll, J., Rumi, M.A., **Brown, C.L.**, Joung, M.C., Moumi, N.A., Ahmed, S., Song, H., Sein, J., Yao, S., Khan, A., Gupta, S., Kulkarni, R., Butt, A., Vikesland, P.J., Pruden, A., Zhang, L. CIWARS: a web server for waterborne antibiotic resistance surveillance using longitudinal metagenomic data. *Journal of Molecular Biology.* https://doi.org/10.1016/j.jmb.2025.169159

14. Talat, A., Bashir, Y., **Brown, C.L.**, Gupta, D., Khaleel, N., Khan, A. Advancing AMR surveillance: metagenomic insights from traditional and UV-enabled advanced wastewater treatment plants. *Environmental Microbiome*. https://link.springer.com/article/10.1186/s40793-024-00658-2
15. Choi, J.M., Manthapuri, V., Keenum, I., **Brown, C.L.**, Xia, K., Vikesland, P., Blair, M., Bott, C., Pruden, A., Zhang, L. A machine learning approach to predicting PPCP removal through various water reuse treatment processes. *RSC Env. Sci.: Water Research and Technology*. https://pubs.rsc.org/en/content/articlehtml/2025/ew/d4ew00892h
16. Choi, J.M., Rumi, M.A., **Brown, C.L.**, Vikesland, P.J., Pruden, A., Zhang, L. ARGfore: a multivariate framework for forecasting antibiotic resistance gene abundances using time-series metagenomic datasets. *IEEE Access*, vol. 14, pp. 34692–34704, 2026. https://doi.org/10.1109/ACCESS.2026.3667074
17. Song, H., Tithi, S., **Brown, C.L.**, Aylward, F., Jensen, R., Zhang, L. VirSeqImprover: an integrated pipeline for viral contig error correction, extension, and annotation. *PeerJ*. https://peerj.com/articles/18515/
18. Rumi, M.A., Oh, M., Davis, B.C., Juvekar, A., **Brown, C.L.**, Vikesland, P.J., Pruden, A., Zhang, L. MetaCompare 2.0: differential ranking of ecological and human health resistome risks. *FEMS*. https://academic.oup.com/femsec/article/100/12/fiae155/7888989
19. Song, Y., Aguilar, D., **Brown, C.L.**, Pruden, A., & Edwards, M. Copper-induced viable-but-not-culturable *Legionella pneumophila* retain infectivity and undergo metabolic shutdown. *Microorganisms*.
20. Kang, S., Maile-Moskowitz, A., **Brown, C.L.**, Pruden, A., Vikesland, P. (2024). Highly multiplexed reverse-transcription loop-mediated isothermal amplification and nanopore sequencing (LAMPore) for wastewater-based surveillance. *Environmental Science & Technology Water*. https://pubs.acs.org/doi/10.1021/acsestwater.3c00690
21. Cullom, A., Spencer, M.S., Williams, M.D., Falkinham III, J.O., **Brown, C.L.**, Edwards, M.A., & Pruden, A. (2023). Premise plumbing pipe materials and in-building disinfectants shape the potential for proliferation of pathogens and antibiotic resistance genes. *Environmental Science & Technology*. https://doi.org/10.1021/acs.est.3c05905
22. Davis, B., **Brown, C.L.**, Liguori, K., Milligan, E., Pruden, A., Keenum, I. Towards standardization of metagenomics for monitoring antibiotic resistance in wastewater and impacted aquatic environments. *Critical Reviews in Environmental Science and Technology*, **53**(19), 1731–1756. https://doi.org/10.1080/10643389.2023.2181620
23. Dai, D., **Brown, C.L.**, Bürgmann, H., Larsson, J., Nambi, I., Zhang, T., Flach, C.-F., Pruden, A., Vikesland, P. Long-read metagenomic sequencing reveals shifts in associations of antibiotic resistance genes with mobile genetic elements from sewage to activated sludge. *Microbiome* **10**, 20 (2022). https://doi.org/10.1186/s40168-021-01216-5
24. Marano, R.B.M., Fernandes, T., Manaia, C.M., Nunes, O., Morrison, D., Berendonk, T.U., Kreuzinger, N., Telson, T., Corno, G., Fatta-Kassinos, D., Merlin, C., Topp, E., Jurkevitch, E., Henn, L., Scott, A., Heß, S., Slipko, K., Laht, M., Kisand, V., ... **Brown, C.L.**, Cytryn, E. (2020). A global multinational survey of cefotaxime-resistant coliforms in urban wastewater treatment plants. *Environment International*, **144**, 106035. https://doi.org/10.1016/j.envint.2020.106035
25. Garner, E., **Brown, C.L.**, Schwake, D.O., Rhoads, W.J., Arango-Argoty, G., Zhang, L., Jospin, G., Coil, D.A., Eisen, J.A., Edwards, M.A., & Pruden, A. (2019). Comparison of whole-genome sequences of *Legionella pneumophila* in tap water and in clinical strains, Flint, Michigan, USA, 2016. *Emerging Infectious Diseases*, **25**(11). https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6810188/

26. Moumi, N., **Brown, C.L.**, Vikesland, P., Pruden, A., Zhang, L. Protein-protein interaction network analysis reveals distinct patterns of antibiotic resistance genes. *IEEE BIBM 2022*, Las Vegas, Nevada, USA.
27. Chandak, S., Zhang, L., **Brown, C.L.**, Huang, L. (2022). Towards automatic curation of antibiotic resistance genes via statement extraction from scientific papers: a benchmark dataset and models. In *Proceedings of the 21st Workshop on Biomedical Language Processing*, pages 402–411, Dublin, Ireland. Association for Computational Linguistics.

In Review

1. Yang, Y., **Brown, C.L.**, Liu, L., Sereika, M., Bygh Nymann Jensen, T., Albertsen, M., Halkjær, P.H., Singleton, C.M. Ecological specificity and interconnectivity in Danish environmental resistomes. *In revision, Nature Communications.* https://www.biorxiv.org/content/10.64898/2026.07.09.737170v1.abstract
2. **Brown, C.L.**[a] and Sorensen, B.[a], Sereika, M., Heidelbach, S., Hugenholtz, P., Albertsen, M., Delogu, F., Nation-scale recovery of complete environmental plasmids pushes the boundaries of plasmid diversity through long-read sequencing *In review.* https://www.biorxiv.org/content/10.64898/2026.10.06.756670v1
3. Rumi MA, Nguyen L, Davis BC, **Brown CL**, Pruden A, Zhang L. Mapping the underlying drivers of resistome risk across diverse environments. doi: 10.21203/rs.3.rs-7085902/v1 *In revision*

[a]Co-first authors

*Corresponding author

Presentations

Major Oral Presentations

11. (invited) Brown, C.L. How should we relate antibiotic resistance indicators in sewage to community resistance? A pilot study of deep, longitudinally sampled sewage metagenomes. *Transatlantic Taskforce on Antibiotic Resistance (TATFAR) meeting.* Mar 2025.
10. (invited) Brown, C.L. Towards metagenomic epidemiology of antibiotic resistant pathogens: delineating pathogen subtypes in sewage and their relation to clinical measures of antibiotic resistance. *Indiana University, Department of Environmental Health.* Feb 2025.
9. (invited) Brown, C.L. Sensitively untangling microbial ecology and evolution through microdiversity-aware metagenomics. *Michigan Technical University, Houghton, MI.* July 2024.
8. (invited) Brown, C.L., Pruden, A. After EDAR6, Surveillance Systems and the Urgency of Access to Wastewater Treatments. *Club Suisse de la presse, Geneva Press Club, preceding WHO AMR Week.* Nov 2022.
7. (invited) Brown, C.L. Harnessing massive biological data to protect natural resources and human health through surveillance, monitoring, and impact assessment. *Invited talk, Oakwood University Graduate Research Seminar.* Sep 2022.
6. Brown, C.L., Nguyen, L., Vikesland, P., Pruden, A., Zhang, L. Capture, diversify, emit: insights into wastewater treatment plants as reservoirs for antibiotic resistance from one year of deep sequencing. *MoBE Gordon Research Conference (GRC) Research Symposium.* 2025.
5. Brown, C.L., Afrin, M.R., McDaniel, L., Maile-Moskowitz, A., Sein, J., Choi, M., Hindi, F., Mullet, J., Emon, M., Moumi, N.A., Blair, M.F., Rao, J., Baffoe-Bonnie, A., Vikesland, P., Pruden, A., Zhang, L. Resistance in sewage associated microbiota mirror patterns of clinical isolates and reflect a variable lagging association with antibiotic usage. *Environmental Dimensions of Antibiotic Resistance (EDAR) 7*, Montreal, Canada, May 2024.
4. Brown, C.L., et al. Resistance in sewage associated microbiota mirror patterns of clinical

isolates and reflect a variable lagging association with antibiotic usage. *AMR, Genomes, Big Data, and Emerging Technologies*, Hinxton, UK, Mar 2024.

3. Brown, C.L., Maile-Moskowitz, A., Vikesland, P., Zhang, L., Pruden, A. Longitudinal metagenomics of wastewater relays signals of community and environmental health that are sensitive to seasonal weather patterns. *Microbial Ecology and Water Engineering Special Section Conference*, Brisbane, Australia, Sep 2023.
2. Brown, C.L., Mullet, J., Maile-Moskowitz, A., Vikesland, P., Zhang, L., Pruden, A. Development and application of mobileOG-db: a unified database and ontology of MGE protein families leveraged for tracking the fate of hospital sewage-derived resistance genes. *Environmental Dimensions of Antibiotic Resistance (EDAR) 6*, Gothenburg, Sweden, Sep 2022.
1. Brown, C.L., Maile-Moskowitz, A., Vikesland, P., Pruden, A., Zhang, L. Metagenomics based environmental monitoring for community and environmental health: preliminary results of a pilot system at three full scale wastewater treatment plants. *Gen. Bio. Comp. Bio. Departmental Seminar.* Sep 2022.

All Other Posters and Presentations (selected)

- Scott, K., Brown, C.L., Song, H., Blair, M., Davis, B., Markham, C., Zhang, L., Pruden, A., Vikesland, P. Evaluating a metagenomic workflow for nontargeted detection and typing of disease agents in wastewater. *AEESP, 2023.*
- Brown, C.L., Rumi, M., Vikesland, P., Pruden, A., Zhang, L. CIWARS: harnessing massive biological data for monitoring antibiotic resistance and pathogens through shotgun metagenomics of wastewater. *CIP-CAR NRT Opening Workshop, 2022.*
- Maile-Moskowitz, A., Gonzalez, R., Finkelstein, C., Brown, C.L., Vikesland, P., Pruden, A. Assessing the impact of the COVID-19 pandemic on antibiotic resistance and the microbial community of wastewater. *2022.*
- Aguilar, D., Song, Y., Brown, C.L., Edwards, M., Pruden, A., Helm, R. Proteomic profiling of a copper resistant *Legionella pneumophila* strain. *2022.*
- Maile-Moskowitz, A., Brown, C.L., Logan, L., Pruden, A., Vikesland, P. Effect of hospital sewage on antibiotic resistance in treated wastewater effluent. *AEESP, 2022.*
- Brown, C.L. Metagenomics-enabled surveillance of the environmental mobilome: towards predictive public health insights. *Virginia Tech GBCB Dept. Seminar, 2021.*
- Brown, C.L., Maile-Moskowitz, A., Snead, D., Logan, L., Vikesland, P., Pruden, A. Persistence of hospital-derived multidrug resistant bacteria in nitrifying sequencing batch reactors. *Wellcome Genome Conference (Online), 2021.*
- Garner, E., Brown, C.L., Schwake, D.O., Rhoads, W.J., Arango-Argoty, G., Zhang, L., Jospin, G., Coil, D., Eisen, J., Edwards, M., Pruden, A. Whole genome sequence comparison of clinical and drinking water *Legionella pneumophila* isolates associated with the Flint Water Crisis. *Microbiology of the Built Environment GRC, 2018.*
- Schwake, O., Brown, C.L., Marr, L. Influence of climate on *Legionella* contamination in automobile washer fluid. *9th International Conference on Legionella, 2017.*
- Brown, C.L., Schwake, O., Pruden, A., Marr, L. Is climate fate? *Legionella* in washer fluid may be a bigger problem in some states than in others. *Fralin Undergraduate Research Fellowship Symposium, 2017.*
- Brown, C.L., Garner, E., Schwake, O., Pruden, A., Edwards, M. Examining the effects of the Flint water crisis on Endemic *Legionella* populations: An ongoing investigation. *Virginia State ASM, 2016.* (oral)

Students Mentored

(on doctoral committee) Thomas Byrne, Virginia Tech GBCB Ph.D. Program

Yat Fei Cheung	Virginia Tech, Computer Modeling and Data Analytics	M.S. Program, Virginia Tech
Chad Kelly	Virginia Tech, Computer Modeling and Data Analytics	Virginia Tech GBCB Ph.D. Program
Loc Nguyen	Virginia Tech, GBCB Ph.D. Program	
Haoqiu Song	Virginia Tech, Computer Science Ph.D. Program	
Nazifa Moumi Ahmed	Virginia Tech, Computer Science Ph.D. Program	
Justin Sein	Virginia Tech, GBCB Ph.D. Program	
Minyoung Choi	Virginia Tech, Biology	Tufts Dental School
Fadi Hindi	Virginia Tech, Biology	U. Miami Medical School
James Mullet*	Virginia Tech, Biology and Biochemistry	MIT [Biology Senior of the Year 2023]
Monica Gurung	Virginia Tech, Civil and Env. Engineering	UCLA, M.S. Program
Callen Barlik	Virginia Tech, Microbiology	Johnson & Johnson
Ana Haines	Virginia Tech, Civil and Env. Engineering	
Delaney Snead**	Virginia Tech, Civil and Env. Engineering	U. Michigan Ph.D. [CEE Senior of Year, 2021]